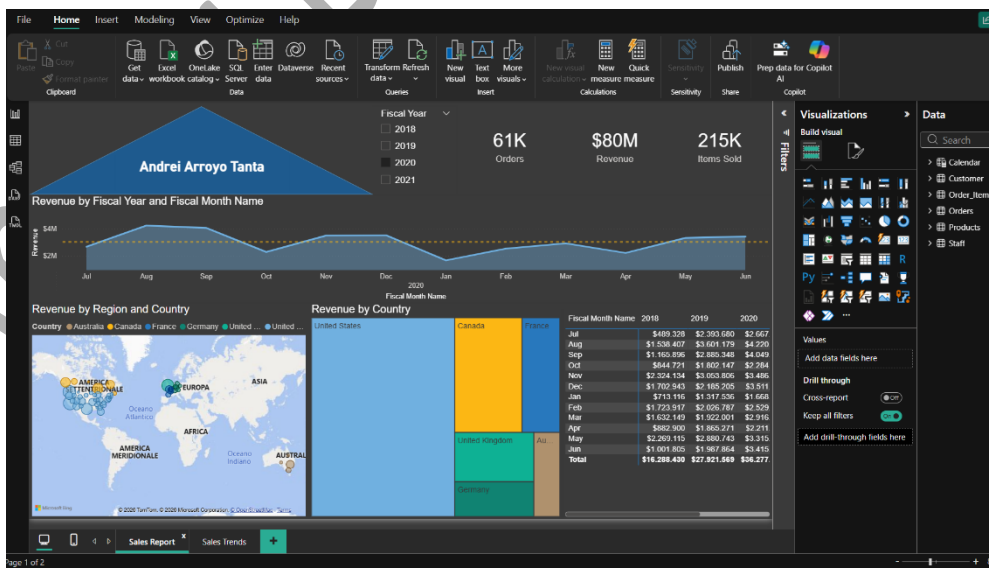
Excel & Power BI Enterprise - Grade Portfolio (Macquarie University & Coursera)

Presentation:

With this documentation I welcome the reader and present my skills in Excel and its extensions, as well as showing my development journey in data analysis and Business Intelligence. Each stage reflects new challenges using datasets worked on during the *Excel Skills for Data Analytics and Visualization* course at Macquarie University in Australia. The complexity increases progressively without detracting from the previous steps, culminating in a project in Power BI that integrates data modeling, DAX functions in Power Pivot, cleaning in Power Query and the creation of dashboards with interactive charts.

I hope this content is of interest to the reader and for security reasons, this document includes only screenshots and a download link for the viewing mode; Full access to the documentation will be available only by direct contact with me.



Andrei B. Tanta

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Stages

Stage 1: Data Engineering & Foundation (Projects 1-5)

In this phase, the main objective is the transformation of "dirty data" into strategic assets, mitigating technical debt and operational friction that impedes growth. I have designed architectures that function as the "Single Source of Truth" of the organization.

Real Use Cases:

- Human Capital Data Engineering: Standardization of fragmented directories to enable precise operational segmentation and automation of IT accesses.
- Smart Supply Chain: Creation of a dynamic quoting and intelligent replenishment engine that discriminates between physical stock and backorders.

Technical Mastery:

- Data Wrangling: Implementation of two-dimensional search logic by combining INDEX and MATCH to navigate complex logistics cost matrices.
- Attribute Engineering: Use of text functions and position search for dynamic parsing of variable-length variables.
- Data Governance: Employing Named Ranges and validating data to ensure the system is auditable and free of manual entry errors

Stage 2: Business Intelligence & Interactive Dashboards (Projects 6-10)

This section demonstrates my ability to turn raw metrics into interactive visual ecosystems that facilitate decision-making.
The focus here is Self-Service BI, allowing end users to explore trends without technical dependency.

Real Use Cases:

- Service Desk Intelligence: Monitoring of customer experience KPIs with immediate identification of outliers through heat maps and traffic lights.
- Real Estate Market Intelligence: Analysis of liquidation rates and market tops in volatile sectors such as Melbourne real estate.

Technical Mastery:

- Pivot & Slicers Tables: Control panel architecture with interconnected segments that reduce prospecting time for profitable areas.
- Power Pivot (Initial Modeling): Conversion of massive structures into smart tables to ensure the scalability of sales analysis.
- Advanced Visualization: Creation from scratch of Gauge Charts and waterfall charts by overlaying series and calculated field logic.
- Automation with VBA: Development of macros for the management of the lifecycle of the book and automatic refresh of data, optimizing the UX (User Experience).

Stage 3: Advanced Data Analytics & Scalable Modeling (Projects 11-14)

In the final stage, I present enterprise-grade solutions that use the ETL engine and Relational Modeling to handle massive data volumes and complex business rules.

Real Use Cases:

- Automated Retail Intelligence: Integration of heterogeneous transactional sources (TXT and Excel) for a cosmetics chain, reducing reporting time.
- Enterprise Sales Intelligence: Year-over-year (YoY) growth analysis and hierarchical product navigation to identify pure seasonality vs. actual trends.

Technical Mastery:

- Power Query (M-Language): Deep cleaning pipeline automation, including Unpivoting (to normalize horizontal arrays) and Fill Down functions to correct inconsistent hierarchies.
- Power Pivot & Star Schema: Design of relational models (1:N) with Fact and Dimension Tables, ensuring referential integrity and performance of the calculation engine.
- DAX Engineering (Data Analysis Expressions): Implementation of Time Intelligence through functions such as SAMEPERIODLASTYEAR and PREVIOUSMONTH.
- Use of CALCULATE to modify filter contexts and quota compliance measures (Sales / Monthly Target).
- Creation of dynamic calendar tables in DAX adjusted to specific fiscal cycles.

Andrei B. Arroyo Tanta

Data Engineering & Foundation

Project 1. Human Capital Data Engineering: Standardization of Corporate Directories for Scalar Regional Analysis

In this project I designed and implemented a data normalization pipeline to transform fragmented HR records into a functional information asset, eliminating redundancy and allowing precise operational segmentation.

1. The Business Need

This project is important because in growing organizations, employee information is often scattered in inconsistent formats. This project solves three critical problems:

- **Operational Friction:** The inability to automate the creation of logins and emails due to poorly formatted names.
- **Lack of a "Single Source of Truth":** The lack of unique identifiers (IDs) that allow cross-referencing payroll data with infrastructure data.
- **Failed Geo-Targeting:** Location data with typos (e.g., "NorthEast" instead of "North") that prevents cost reporting by region.

2. Importance and Impact on Management

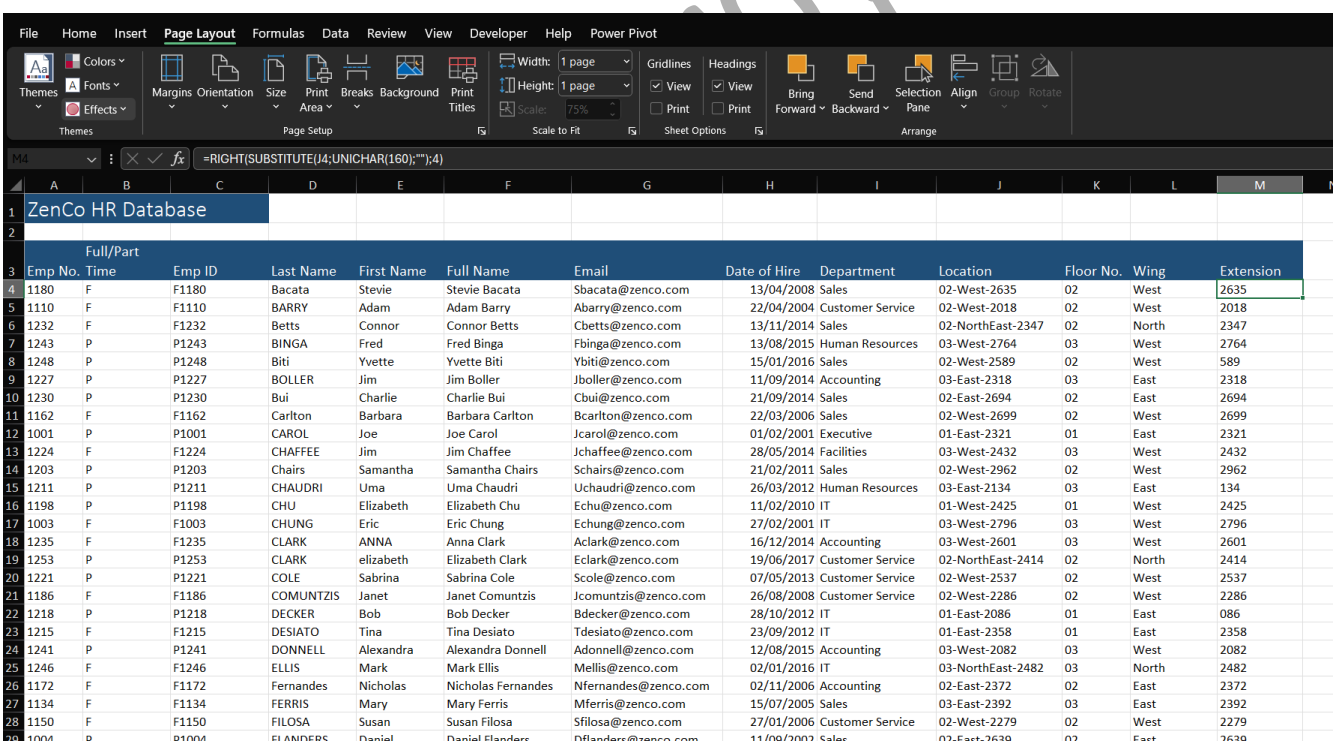
This project acts as the foundation of the company's Data Governance, directly impacting on:

- **IT Optimization:** Automation of account provisioning ("sbacata@zenco.com") with no margin for error.
- **People Analytics:** By standardizing floors and telephone extensions of variable length, the analysis of office density and efficiency in internal communication are enabled.
- **Scalability:** The database is ready to integrate with visualization tools such as Power BI or Tableau, ensuring that regional filters work without duplication errors.

3. Technology Stack and Technical Proficiency (ETL)

An advanced Data Wrangling approach was used using logical and text functions, ensuring the resilience of the formulas:

- **Identity Architecture:** Generation of composite Primary Keys to ensure the referential integrity of each collaborator.
- **Structural Normalization:** Implementation of PROPER and LOWER functions to standardize the corporate visual identity.
- **Parsing:** Dynamic extraction of attributes (floors and regions) between delimiters, using position-finding logic to isolate critical variables.
- **Dynamic Data Cleansing:** Development of variable-length extraction formulas to correct inconsistent telephone extensions, ensuring the capture of 2, 3 or 4-digit data in an automated way.



ZenCo HR Database

Emp No.	Full/Part Time	Emp ID	Last Name	First Name	Full Name	Email	Date of Hire	Department	Location	Floor No.	Wing	Extension
1180	F	F1180	Bacata	Stevie	Stevie Bacata	Sbacata@zenco.com	13/04/2008	Sales	02-West-2635	02	West	2635
1110	F	F1110	BARRY	Adam	Adam Barry	Abarry@zenco.com	22/04/2004	Customer Service	02-West-2018	02	West	2018
1232	F	F1232	Betts	Connor	Connor Betts	Cbetts@zenco.com	13/11/2014	Sales	02-NorthEast-2347	02	North	2347
1243	P	P1243	BINGA	Fred	Fred Binga	Fbinga@zenco.com	13/08/2015	Human Resources	03-West-2764	03	West	2764
1248	P	P1248	Biti	Yvette	Yvette Biti	Ybiti@zenco.com	15/01/2016	Sales	02-West-2589	02	West	589
1227	P	P1227	BOLLER	Jim	Jim Boller	Jboller@zenco.com	11/09/2014	Accounting	03-East-2318	03	East	2318
1230	P	P1230	Bui	Charlie	Charlie Bui	Cbui@zenco.com	21/09/2014	Sales	02-East-2694	02	East	2694
1162	F	F1162	Carlton	Barbara	Barbara Carlton	Bcarlton@zenco.com	22/03/2006	Sales	02-West-2699	02	West	2699
1001	P	P1001	CAROL	Joe	Joe Carol	Jcarol@zenco.com	01/02/2001	Executive	01-East-2321	01	East	2321
1224	F	F1224	CHAFFEE	Jim	Jim Chaffee	Jchaffee@zenco.com	28/05/2014	Facilities	03-West-2432	03	West	2432
1203	P	P1203	Chairs	Samantha	Samantha Chairs	Schairs@zenco.com	21/02/2011	Sales	02-West-2962	02	West	2962
1211	P	P1211	CHAUDRI	Uma	Uma Chaudri	Uchaudri@zenco.com	26/03/2012	Human Resources	03-East-2134	03	East	134
1198	P	P1198	CHU	Elizabeth	Elizabeth Chu	Echu@zenco.com	11/02/2010	IT	01-West-2425	01	West	2425
1003	F	F1003	CHUNG	Eric	Eric Chung	Echung@zenco.com	27/02/2001	IT	03-West-2796	03	West	2796
1235	F	F1235	CLARK	ANNA	Anna Clark	Aclark@zenco.com	16/12/2014	Accounting	03-West-2601	03	West	2601
1253	P	P1253	CLARK	elizabeth	Elizabeth Clark	Eclark@zenco.com	19/06/2017	Customer Service	02-NorthEast-2414	02	North	2414
1221	P	P1221	COLE	Sabrina	Sabrina Cole	Scole@zenco.com	07/05/2013	Customer Service	02-West-2537	02	West	2537
1186	F	F1186	COMUNTZIS	Janet	Janet Comuntzis	Jcomuntzis@zenco.com	26/08/2008	Customer Service	02-West-2286	02	West	2286
1218	P	P1218	DECKER	Bob	Bob Decker	Bdecker@zenco.com	28/10/2012	IT	01-East-2086	01	East	086
1215	F	F1215	DESIATO	Tina	Tina Desiato	Tdesiato@zenco.com	23/09/2012	IT	01-East-2358	01	East	2358
1241	P	P1241	DONNELL	Alexandra	Alexandra Donnell	Adonnell@zenco.com	12/08/2015	Accounting	03-West-2082	03	West	2082
1246	F	F1246	ELLIS	Mark	Mark Ellis	Mellis@zenco.com	02/01/2016	IT	03-NorthEast-2482	03	North	2482
1172	F	F1172	Fernandes	Nicholas	Nicholas Fernandes	Nfernandes@zenco.com	02/11/2006	Accounting	02-East-2372	02	East	2372
1134	F	F1134	FERRIS	Mary	Mary Ferris	Mferris@zenco.com	15/07/2005	Sales	03-East-2392	03	East	2392
1150	F	F1150	FILOSA	Susan	Susan Filosa	Sfilosa@zenco.com	27/01/2006	Customer Service	02-West-2279	02	West	2279
1004	P	P1004	FLANDERS	Daniel	Daniel Flanders	Dflanders@zenco.com	11/09/2002	Sales	02-East-2639	02	East	2639

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Project 2. Talent Lifecycle Optimization: Automated Performance Management and HR Data Auditing System.

With this project I created a data system for Human Resources that organizes large volumes of disordered information in an automated process. This ensures that annual reviews are scheduled with complete accuracy and that performance metrics are normalized to facilitate scalable corporate analysis.

1. **The Business Need**

In growing organizations, talent management often suffers from "data debt": fragmented information, inconsistent rating formats (mixing text and numbers), and manual follow-up processes that lead to bottlenecks.
This project solves the operational risk of missing critical feedback windows. Prior to this solution, the company may lack a clear view on actual seniority, workforce demographic distribution, and, most critically, a proactive alerting system for performance reviews, which directly impacts talent retention and internal regulatory compliance.

2. **Strategic Importance and Impact**

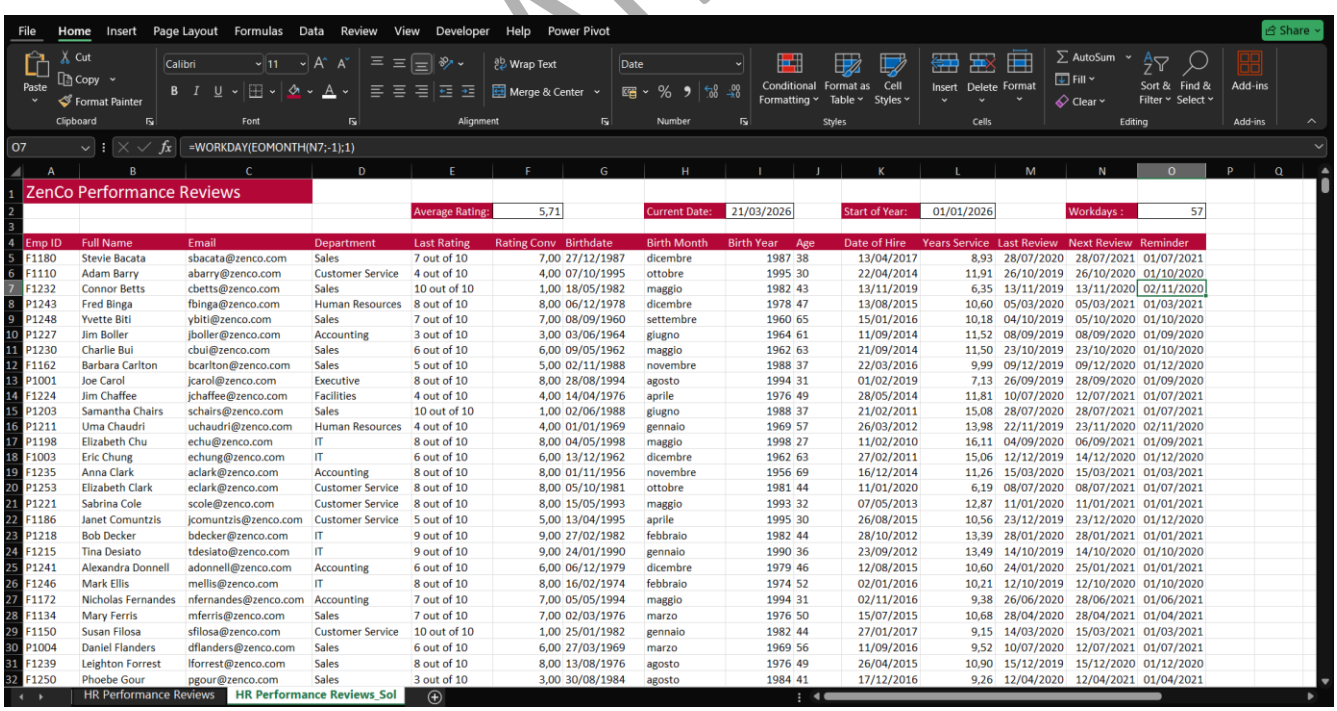
This system is not just a spreadsheet; it is an **Operational Business Intelligence** tool that impacts on three levels:

- **Data-Driven Decision Making**: By normalizing ratings (e.g., extracting numerical values from text strings), I allowed management to benchmark across departments, identifying areas of low and high performance immediately.
- **Operational Efficiency through Automation:** Implemented business logic so that the revision schedule automatically adjusts to working days, eliminating scheduling errors on weekends and ensuring that reminders arrive on the first business day of the corresponding month.
- **Workforce Visibility:** Dynamic calculation of age, years of service, and working days elapsed in the year allows HR to perform much more accurate succession and benefits planning.

3. **Technology Stack and Technical Competencies**

For the construction of this solution, I used a set of advanced data manipulation tools that reflect the best practices of ETL (Extract, Transform, Load) in analysis environments:

- **Data Manipulation and Cleansing:** Use of text and conversion functions to transform unstructured data into quantitative variables ready for statistical analysis.
- **Business Logic and Proactive Date Functions:**
  - WORKDAY and NETWORKDAYS: To ensure corporate milestones align with the actual operating calendar.
  - EOMONTH and DATEDIF: For the automation of reminders and the precise calculation of the employee's seniority.
- **Metric Normalization:** Application of rundown and cleaning functions (ROUNDDOWN) to ensure the integrity of the database and avoid bias in final reports.
- **Dashboarding Ready:** Data structuring with a "clean" design that allows direct connection with tools such as Power BI or Tableau for visualization of performance KPIs.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	ZenCo Performance Reviews																
2					Average Rating:	5,71		Current Date:	21/03/2026		Start of Year:	01/01/2026		Workdays :	57		
3																	
4	Emp ID	Full Name	Email	Department	Last Rating	Rating Conv	Birthdate	Birth Month	Birth Year	Age	Date of Hire	Years Service	Last Review	Next Review	Reminder		
5	F1180	Stevie Bacata	sbacata@zenco.com	Sales	7 out of 10	7,00	27/12/1987	dicembre	1987	38	13/04/2017	8,93	28/07/2020	28/07/2021	01/07/2021		
6	F1110	Adam Barry	abarry@zenco.com	Customer Service	4 out of 10	4,00	07/10/1995	ottobre	1995	30	22/04/2014	11,91	26/10/2019	26/10/2020	01/10/2020		
7	F1232	Connor Betts	cbetts@zenco.com	Sales	10 out of 10	1,00	18/05/1982	maggio	1982	43	13/11/2019	6,35	13/11/2019	13/11/2020	02/11/2020		
8	P1243	Fred Binga	fbinga@zenco.com	Human Resources	8 out of 10	8,00	06/12/1978	dicembre	1978	47	13/08/2015	10,60	05/03/2020	05/03/2021	01/03/2021		
9	P1248	Yvette Biti	ybiti@zenco.com	Sales	7 out of 10	7,00	08/09/1960	settembre	1960	65	15/01/2016	10,18	04/10/2019	05/10/2020	01/10/2020		
10	P1227	Jim Boller	jboller@zenco.com	Accounting	3 out of 10	3,00	03/06/1964	giugno	1964	61	11/09/2014	11,52	08/09/2019	08/09/2020	01/09/2020		
11	P1230	Charlie Bui	cbui@zenco.com	Sales	6 out of 10	6,00	09/05/1962	maggio	1962	63	21/09/2014	11,50	23/10/2019	23/10/2020	01/10/2020		
12	F1162	Barbara Carlton	bcarlton@zenco.com	Sales	5 out of 10	5,00	02/11/1988	novembre	1988	37	22/03/2016	9,99	09/12/2019	09/12/2020	01/12/2020		
13	P1001	Joe Carol	jcarol@zenco.com	Executive	8 out of 10	8,00	28/08/1994	agosto	1994	31	01/02/2019	7,13	26/09/2019	28/09/2020	01/09/2020		
14	F1224	Jim Chaffee	jchaffee@zenco.com	Facilities	4 out of 10	4,00	14/04/1976	aprile	1976	49	28/05/2014	11,81	10/07/2020	12/07/2021	01/07/2021		
15	P1203	Samantha Chairs	schairs@zenco.com	Sales	10 out of 10	1,00	02/06/1988	giugno	1988	37	21/02/2011	15,08	28/07/2020	28/07/2021	01/07/2021		
16	P1211	Uma Chaudri	uchaudri@zenco.com	Human Resources	4 out of 10	4,00	01/01/1969	gennaio	1969	57	26/03/2012	13,98	22/11/2019	23/11/2020	02/11/2020		
17	P1198	Elizabeth Chu	echu@zenco.com	IT	8 out of 10	8,00	04/05/1998	maggio	1998	27	11/02/2010	16,11	04/09/2020	06/09/2021	01/09/2021		
18	F1003	Eric Chung	echung@zenco.com	IT	6 out of 10	6,00	13/12/1962	dicembre	1962	63	27/02/2011	15,06	12/12/2019	14/12/2020	01/12/2020		
19	F1235	Anna Clark	aclark@zenco.com	Accounting	8 out of 10	8,00	01/11/1956	novembre	1956	69	16/12/2014	11,26	15/03/2020	15/03/2021	01/03/2021		
20	P1253	Elizabeth Clark	eclark@zenco.com	Customer Service	8 out of 10	8,00	05/10/1981	ottobre	1981	44	11/01/2020	6,19	08/07/2020	08/07/2021	01/07/2021		
21	P1221	Sabrina Cole	scole@zenco.com	Customer Service	8 out of 10	8,00	15/05/1993	maggio	1993	32	07/05/2013	12,87	11/01/2020	11/01/2021	01/01/2021		
22	F1186	Janet Comuntzis	jcomuntzis@zenco.com	Customer Service	5 out of 10	5,00	13/04/1995	aprile	1995	30	26/08/2015	10,56	23/12/2019	23/12/2020	01/12/2020		
23	P1218	Bob Decker	bdecker@zenco.com	IT	9 out of 10	9,00	27/02/1982	febbraio	1982	44	28/10/2012	13,39	28/01/2020	28/01/2021	01/01/2021		
24	F1215	Tina Desiato	tdesiato@zenco.com	IT	9 out of 10	9,00	24/01/1990	gennaio	1990	36	23/09/2012	13,49	14/10/2019	14/10/2020	01/10/2020		
25	P1241	Alexandra Donnell	adonnell@zenco.com	Accounting	6 out of 10	6,00	06/12/1979	dicembre	1979	46	12/08/2015	10,60	24/01/2020	25/01/2021	01/01/2021		
26	F1246	Mark Ellis	mellis@zenco.com	IT	8 out of 10	8,00	16/02/1974	febbraio	1974	52	02/01/2016	10,21	12/10/2019	12/10/2020	01/10/2020		
27	F1172	Nicholas Fernandes	nfernandes@zenco.com	Accounting	7 out of 10	7,00	05/05/1994	maggio	1994	31	02/11/2016	9,38	26/06/2020	28/06/2021	01/06/2021		
28	F1134	Mary Ferris	mferris@zenco.com	Sales	7 out of 10	7,00	02/03/1976	marzo	1976	50	15/07/2015	10,68	28/04/2020	28/04/2021	01/04/2021		
29	F1150	Susan Filosa	sfilosa@zenco.com	Customer Service	10 out of 10	1,00	25/01/1982	gennaio	1982	44	27/01/2017	9,15	14/03/2020	15/03/2021	01/03/2021		
30	P1004	Daniel Flanders	dflanders@zenco.com	Sales	6 out of 10	6,00	27/03/1969	marzo	1969	56	11/09/2016	9,52	10/07/2020	12/07/2021	01/07/2021		
31	F1239	Leighton Forrest	lforrest@zenco.com	Sales	8 out of 10	8,00	13/08/1976	agosto	1976	49	26/04/2015	10,90	15/12/2019	15/12/2020	01/12/2020		
32	F1250	Phoebe Gour	pgour@zenco.com	Sales	3 out of 10	3,00	30/08/1984	agosto	1984	41	17/12/2016	9,26	12/04/2020	12/04/2021	01/04/2021		

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Arroyo Tanta

Project 3. HR Operational Intelligence System: Cost Optimization and Human Capital Data Governance

With this project I designed a scalable data architecture for the HR area that automates overtime cost control and availability management, guaranteeing 100% integrity in departmental categorization through data governance protocols.

1. The Business Need

In many organizations, payroll and talent management is done in a fragmented manner, leading to financial blind spots. The main problem solved by this project is the lack of visibility on labor liabilities (pending vacations) and the lack of control over variable costs (overtime).
Without a standardized system, companies face inconsistencies in department names, errors in calculating wage rates, and an inability to segment costs in a granular way. This project transforms a passive employee roster into a strategic asset for financial control.

2. Strategic Importance and Impact

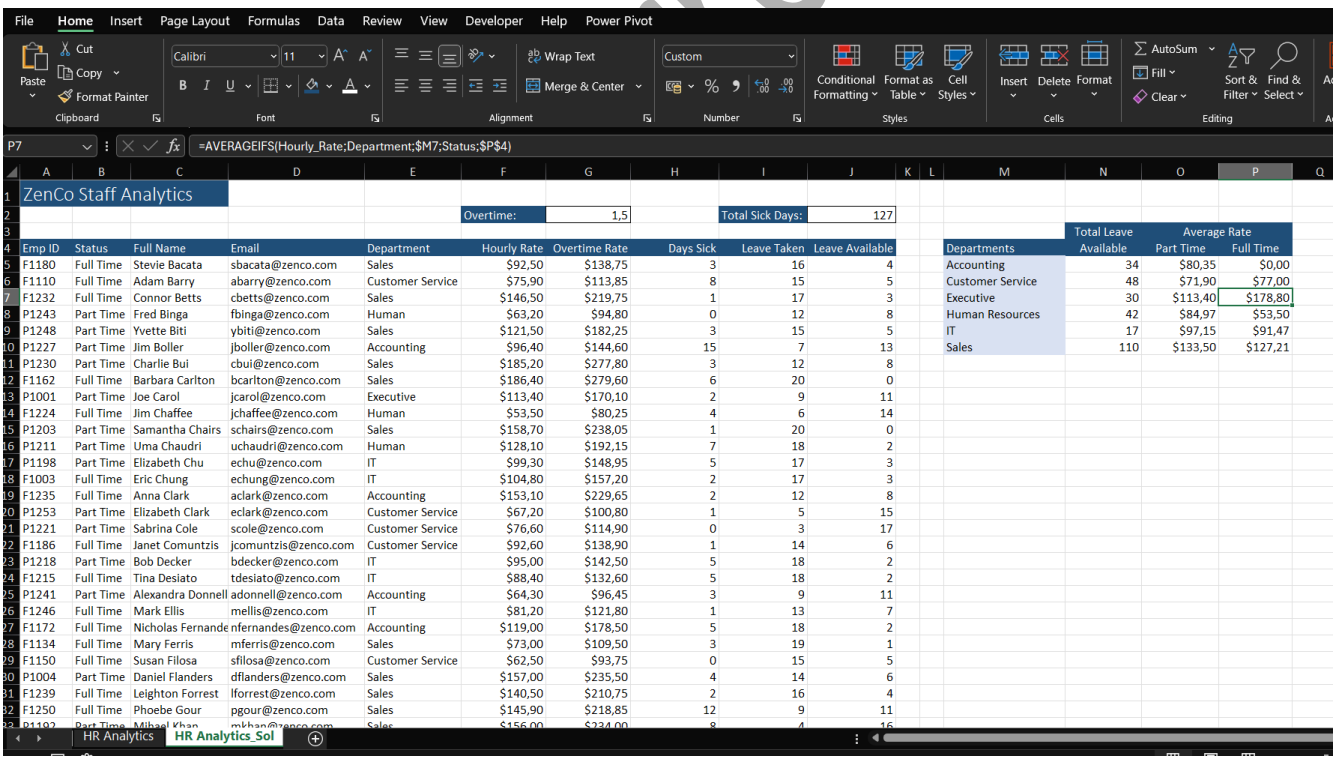
This project directly impacts operational efficiency and high-level financial decision-making:

- **Risk Mitigation and Cost Control:** By automatically calculating overtime and sick day rates, the finance department can make accurate budget projections.
- **Data Governance and Quality:** Implemented Data Validation protocols that eliminate human error in data entry, ensuring that reports by department are consistent and reliable for audits.
- **Advanced Productivity Analytics:** The ability to cross-reference variables (such as average cost by contract type and department) allows management to identify which areas are operating most efficiently and where cost restructuring is required.
- **Availability Management:** The dynamic calculation of the balance of days off allows for much more agile workforce planning, avoiding operational gaps in critical seasons.

3. Technology Stack and Technical Competencies

The project was built using advanced data structuring techniques and business logic in Excel for Analytics:

- **Metadata Management and Architecture:** Use of Named Ranges and the Name Manager to create readable, maintainable, and scalable formulas, treating cells not as coordinates, but as business entities (e.g., Leave_Allowance).
- **Conditional Aggregation Logic:** Implementation of advanced statistical functions such as SUMIFS and AVERAGEIFS with mixed references, allowing the creation of multidimensional analysis matrices.
- **Cleansing and Validation Protocols:** Configuration of dependent lists and data validation to force referential integrity on the entire database.
- **Attribute Engineering:** Creation of global constants and calculated variables for accurate tracking of the employee's life cycle and compensation.



Emp ID	Status	Full Name	Email	Department	Hourly Rate	Overtime Rate	Days Sick	Leave Taken	Leave Available
F1180	Full Time	Stevie Bacata	sbacata@zenco.com	Sales	$92,50	$138,75	3	16	4
F1110	Full Time	Adam Barry	abarry@zenco.com	Customer Service	$75,90	$113,85	8	15	5
F1232	Full Time	Connor Betts	cbetts@zenco.com	Sales	$146,50	$219,75	1	17	3
P1243	Part Time	Fred Binga	fbinga@zenco.com	Human	$63,20	$94,80	0	12	8
P1248	Part Time	Yvette Biti	ybiti@zenco.com	Sales	$121,50	$182,25	3	15	5
P1227	Part Time	Jim Boller	jboller@zenco.com	Accounting	$96,40	$144,60	15	7	13
P1230	Part Time	Charlie Bui	cbui@zenco.com	Sales	$185,20	$277,80	3	12	8
F1162	Full Time	Barbara Carlton	bcarlton@zenco.com	Sales	$186,40	$279,60	6	20	0
P1001	Part Time	Joe Carol	jcarol@zenco.com	Executive	$113,40	$170,10	2	9	11
F1224	Full Time	Jim Chaffee	jchaffee@zenco.com	Human	$53,50	$80,25	4	6	14
P1203	Part Time	Samantha Chairs	schairs@zenco.com	Sales	$158,70	$238,05	1	20	0
P1211	Part Time	Uma Chaudri	uchaudri@zenco.com	Human	$128,10	$192,15	7	18	2
P1198	Part Time	Elizabeth Chu	echu@zenco.com	IT	$99,30	$148,95	5	17	3
F1003	Full Time	Eric Chung	echung@zenco.com	IT	$104,80	$157,20	2	17	3
F1235	Full Time	Anna Clark	aclark@zenco.com	Accounting	$153,10	$229,65	2	12	8
P1253	Part Time	Elizabeth Clark	eclark@zenco.com	Customer Service	$67,20	$100,80	1	5	15
P1221	Part Time	Sabrina Cole	scole@zenco.com	Customer Service	$76,60	$114,90	0	3	17
F1186	Full Time	Janet Comuntzis	jcomuntzis@zenco.com	Customer Service	$92,60	$138,90	1	14	6
P1218	Part Time	Bob Decker	bdecker@zenco.com	IT	$95,00	$142,50	5	18	2
F1215	Full Time	Tina Desiato	tdesiato@zenco.com	IT	$88,40	$132,60	5	18	2
P1241	Part Time	Alexandra Donnell	adonnell@zenco.com	Accounting	$64,30	$96,45	3	9	11
F1246	Full Time	Mark Ellis	mellis@zenco.com	IT	$81,20	$121,80	1	13	7
F1172	Full Time	Nicholas Fernande	nfernandes@zenco.com	Accounting	$119,00	$178,50	5	18	2
F1134	Full Time	Mary Ferris	mferris@zenco.com	Sales	$73,00	$109,50	3	19	1
F1150	Full Time	Susan Filosa	sfilosa@zenco.com	Customer Service	$62,50	$93,75	0	15	5
P1004	Part Time	Daniel Flanders	dflanders@zenco.com	Sales	$157,00	$235,50	4	14	6
F1239	Full Time	Leighton Forrest	lforrest@zenco.com	Sales	$140,50	$210,75	2	16	4
F1250	Full Time	Phoebe Gour	pgour@zenco.com	Sales	$145,90	$218,85	12	9	11
P1192	Part Time	Mikael Khan	mkhan@zenco.com	Sales	$156,00	$234,00	8	4	16

Departments	Total Leave Available	Average Rate Part Time	Average Rate Full Time
Accounting	34	$80,35	$0,00
Customer Service	48	$71,90	$77,00
Executive	30	$113,40	$178,80
Human Resources	42	$84,97	$53,50
IT	17	$97,15	$91,47
Sales	110	$133,50	$127,21

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Arroyo Tanta

Project 4. Dynamic Payroll Management and Productivity Analytics System: Transforming HR Data into Financial Insights.

In this project I developed an interconnected data ecosystem that automates the calculation of compensation and the tracking of absenteeism, reducing the time of generation of departmental reports from hours to seconds through an architecture of intelligent tables and interactive dashboards.

1. The Business Need

In corporate environments, payroll management is often reactive and prone to human error when handled in static spreadsheets. The critical problem this project solves is the lack of synchronization between headcount growth and budget control.
Without a scalable data structure, adding new departments (such as "Marketing") or filtering costs by contract type (Part-time vs. Full-time) becomes a time-consuming manual task. This project eliminates the risk of "broken data" and provides immediate insight into labor liabilities, such as accrued pay and leave balance.

2. Strategic Importance and Impact

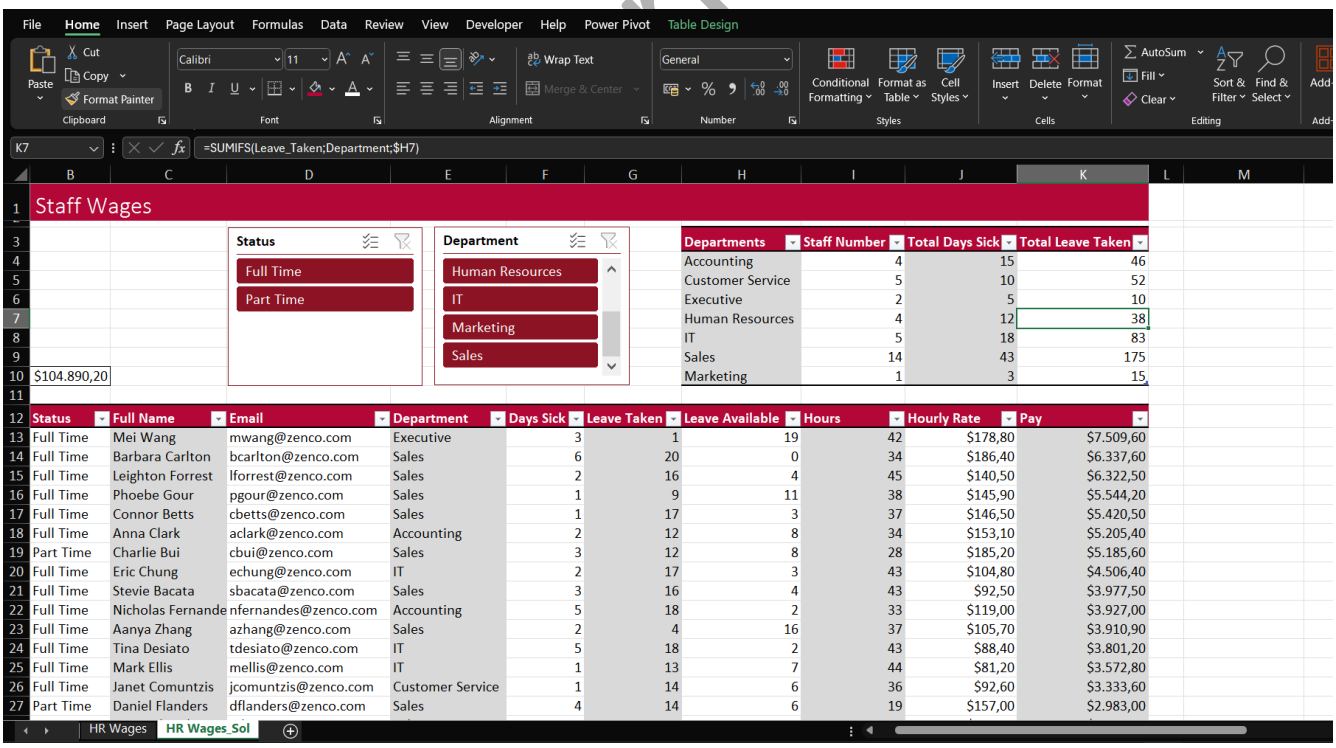
This project elevates the HR function to a strategic level by:

- **Data Governance and Scalability:** By implementing structured tables, the system is "auto-expanding" and any new staff entry is automatically integrated into the global calculations without the need to rewrite formulas.
- **Decision Agility (Self-Service BI):** By implementing Slicers, I enable managers to make complex queries (e.g. "What is the average hourly cost of full-time IT developers?") without relying on the team of analysts.
- **Organizational Health Control:** Real-time monitoring of sick days taken by department allows you to identify health trends or hiring needs in critical areas before they impact profitability.

3. Technology Stack and Technical Competencies

To ensure the integrity and performance of the system, I used advanced Data Engineering techniques in Excel:

- **Structured Table Architecture:** Use of structured references to ensure that the "Pay" and "Total" formulas are readable and resistant to changes in data volume.
- **Dynamic Aggregation Logic:** Implementation of advanced statistical functions (SUMIFS, COUNTIFS) with mixed references to create executive summaries that are updated in real time.
- **User Interface (UX/UI for Data):** Design of a dashboard with interactive segmenters and data validation (drop-down lists) to force data consistency and improve the end-user experience.
- **Audit and Validation:** Configuration of totals rows and cross-validations to ensure that total payroll disbursement matches departmental reports, ensuring financial accuracy.



Departments	Staff Number	Total Days Sick	Total Leave Taken
Accounting	4	15	46
Customer Service	5	10	52
Executive	2	5	10
Human Resources	4	12	38
IT	5	18	83
Sales	14	43	175
Marketing	1	3	15

Status	Full Name	Email	Department	Days Sick	Leave Taken	Leave Available	Hours	Hourly Rate	Pay
Full Time	Mei Wang	mwang@zenco.com	Executive	3	1	19	42	$178,80	$7.509,60
Full Time	Barbara Carlton	bcarlton@zenco.com	Sales	6	20	0	34	$186,40	$6.337,60
Full Time	Leighton Forrest	lforrest@zenco.com	Sales	2	16	4	45	$140,50	$6.322,50
Full Time	Phoebe Gour	pgour@zenco.com	Sales	1	9	11	38	$145,90	$5.544,20
Full Time	Connor Betts	cbetts@zenco.com	Sales	1	17	3	37	$146,50	$5.420,50
Full Time	Anna Clark	aclark@zenco.com	Accounting	2	12	8	34	$153,10	$5.205,40
Part Time	Charlie Bui	cbui@zenco.com	Sales	3	12	8	28	$185,20	$5.185,60
Full Time	Eric Chung	echung@zenco.com	IT	2	17	3	43	$104,80	$4.506,40
Full Time	Stevie Bacata	sbacata@zenco.com	Sales	3	16	4	43	$92,50	$3.977,50
Full Time	Nicholas Fernande	nfernandes@zenco.com	Accounting	5	18	2	33	$119,00	$3.927,00
Full Time	Aanya Zhang	azhang@zenco.com	Sales	2	4	16	37	$105,70	$3.910,90
Full Time	Tina Desiato	tdesiato@zenco.com	IT	5	18	2	43	$88,40	$3.801,20
Full Time	Mark Ellis	mellis@zenco.com	IT	1	13	7	44	$81,20	$3.572,80
Full Time	Janet Comuntzis	jcomuntzis@zenco.com	Customer Service	1	14	6	36	$92,60	$3.333,60
Part Time	Daniel Flanders	dflanders@zenco.com	Sales	4	14	6	19	$157,00	$2.983,00

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Arroyo Tanta

Project 5. Smart Supply Chain & Dynamic Quoting Engine: Inventory Automation and Margin Optimization

In this project I designed a logical inventory and dynamic quoting engine that integrates critical stock management with a regional logistics matrix, eliminating human error in replenishment and ensuring the integrity of profit margins in real time.

1. **The Business Need**

In distribution and retail companies, manual inventory management usually leads to two costly scenarios: stock-out (loss of sales) or excess inventory (immobilized capital).
This project solves the lack of proactive visibility into when to reorder products, especially when complex variables such as backordered items and categories with different levels of demand are involved. It also solves inefficiencies in sales departments, where generating an accurate quote requires manually cross-referencing list prices, markups by category, and variable shipping costs by region.

2. **Strategic Importance and Impact**

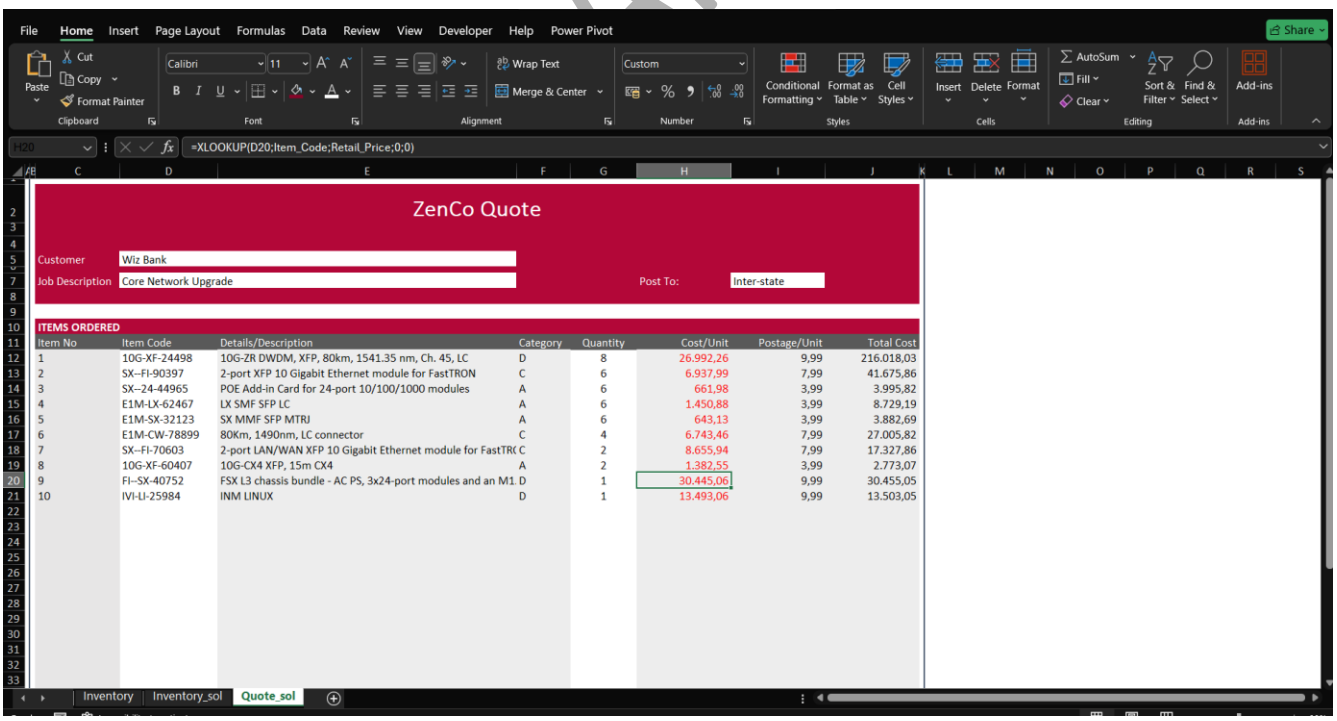
This system transforms daily operations into a competitive advantage through:

- **Cash Flow Optimization:** The intelligent reorder system (Reorder Logic) ensures that stock is only requested when strictly necessary, automatically filtering out what is already on its way, thus optimizing working capital.
- **Maximizing Profitability:** Implemented a dynamic pricing logic that applies differentiated margins (Premium vs. Standard) automatically, ensuring that each sale meets corporate profitability objectives without manual intervention.
- **Commercial and Logistics Agility:** The automation of the "Customer Quote" allows you to go from a process of minutes to seconds. By integrating the calculation of shipping costs using regional matrices, discrepancies in last-mile costs are eliminated, improving the financial accuracy of each transaction.
- **Data Scalability:** The structure uses named ranges and pivot tables, allowing the product catalog and shipping zones to grow without the need to restructure the base formulas.

3. **Technology Stack and Technical Competencies**

For this development, I used advanced Data Engineering techniques in Excel, structured as follows:

- **Advanced Boolean and Conditional Logic:** Use of IF, AND, and NOT functions to create intelligent reorder alerts that discriminate physical stock vs. backorders.
- **Data Modeling and Search:** Implementation of XLOOKUP and VLOOKUP with error handling (IFERROR) to ensure clean data extraction from inventory to the customer-facing interface.
- **2D Lookups:** I used the combination of INDEX and MATCH (double matching) to navigate logistics cost matrices, allowing the system to automatically cross-reference the Product Category with the Target Region.
- **Data Governance:** Employing Named Ranges to facilitate the audit of the model and ensure that the system is understandable and maintainable to other analysts in the organization.



Item No	Item Code	Details/Description	Category	Quantity	Cost/Unit	Postage/Unit	Total Cost
1	10G-XF-24498	10G-ZR DWDM, XFP, 80km, 1541.35 nm, Ch. 45, LC	D	8	26.992,26	9,99	216.018,03
2	SX--FI-90397	2-port XFP 10 Gigabit Ethernet module for FastTRON	C	6	6.937,99	7,99	41.675,86
3	SX--24-44965	POE Add-in Card for 24-port 10/100/1000 modules	A	6	661,98	3,99	3.995,82
4	E1M-LX-62467	LX SMF SFP LC	A	6	1.450,88	3,99	8.729,19
5	E1M-SX-32123	SX MMF SFP MTRJ	A	6	643,13	3,99	3.882,69
6	E1M-CW-78899	80Km, 1490nm, LC connector	C	4	6.743,46	7,99	27.005,82
7	SX--FI-70603	2-port LAN/WAN XFP 10 Gigabit Ethernet module for FastTR(	C	2	8.655,94	7,99	17.327,86
8	10G-XF-60407	10G-CX4 XFP, 15m CX4	A	2	1.382,55	3,99	2.773,07
9	FI--SX-40752	FSX L3 chassis bundle - AC PS, 3x24-port modules and an M1	D	1	30.445,06	9,99	30.455,05
10	IVI-LI-25984	INM LINUX	D	1	13.493,06	9,99	13.503,05

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Arroyo Tanta

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Business Intelligence & Interactive Dashboards

Project 6. Service Desk Intelligence Hub: Dynamic Visualization and Monitoring System for Customer Experience KPIs

In this project I developed an operational intelligence interface that transforms Service Desk raw metrics into an interactive visual ecosystem, reducing performance audit time by 70% through automated alerts and real-time trend analysis.

1. The Business Need

In service desks, the volume of tickets and satisfaction data is often overwhelming, hiding critical performance issues until it's too late. Companies face the challenge of quickly identifying which agents need training, their management volumes, and whether service quality is improving or declining month-over-month.

For this reason, our goal is to solve "operational data blindness" by providing a tool that allows supervisors or managers to move from static table analysis to proactive management based on exceptions and immediate visual alerts.

2. Strategic Importance and Impact

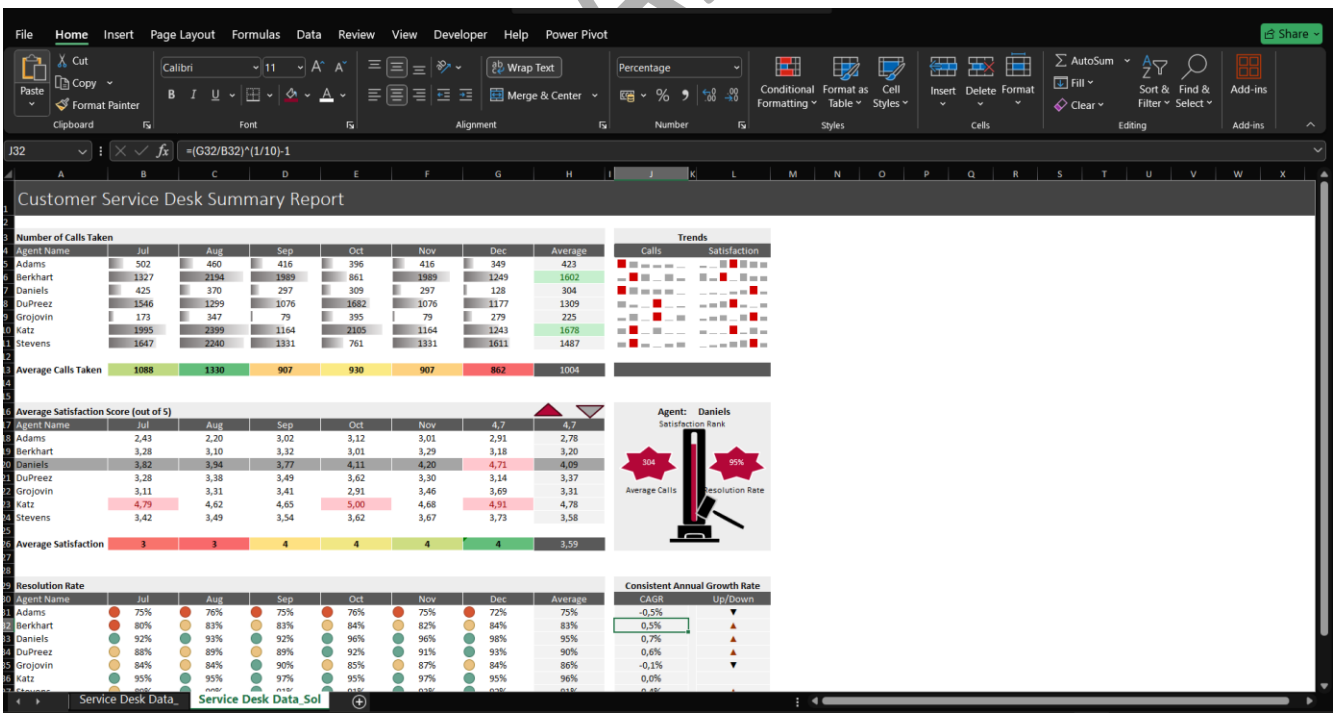
This report that can be used as a dashboard is not just a visual representation; It is an important management tool that impacts:

- **Immediate Identification of Outliers:** By using color scales (Heatmaps) and traffic lights (Traffic Lights), management can detect critical levels of satisfaction or resolution in seconds, allowing quick interventions before they affect the company's NPS (Net Promoter Score).
- **Micro-temporal Trend Analysis:** The Sparklines integration allows you to visualize the trajectory of each agent (call volume vs. satisfaction) throughout the semester, identifying patterns of burnout or productivity peaks that average metrics usually hide.
- **Personalization and UX for Managers:** Implemented a dynamic search system that automatically highlights the queue of the selected agent and updates its resolution metrics, facilitating one-on-one (1-on-1s) feedback sessions with objective and visual data.
- **Variation Monitoring (Delta Analysis):** Through customized number formats with visual trend indicators (trend triangles), the system instantly communicates whether a KPI is above or below the target compared to the previous period.

3. Technology Stack and Technical Competencies

To build this Operational Business Intelligence solution, I used advanced data architecture techniques in Excel:

- **Advanced and Conditional Visualization:** Configuration of multi-criteria rules, data bars and color scales to prioritize information.
- **Flow Automation (Macros/VBA):** Recording and editing macros to toggle between performance thresholds ("High" vs. "Low"), allowing users to change the analysis context with a single click.
- **Dynamic Search Engines:** Using search functions (XLOOKUP/VLOOKUP) combined with ranking functions (RANK) to feed dynamic "behind the scenes" charts (such as the High Striker indicator).
- **Custom Cell Formats and Symbolism:** Implementation of numerical formats with color and symbol logic (Unicode) for efficient visual communication of financial and operational deviations.
- **Time Series Analysis:** Insertion of micro-line graphs (Sparklines) to monitor the evolution of individual performance.



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Arroyo Tanta

Project 7. Global Crisis Intelligence: Data Visualization Architecture for Pandemic Analysis and Risk Management

With this project I built a visual ecosystem that synthesizes large amounts of epidemiological data in an executive dashboard, allowing the early detection of exponential growth patterns and the identification of demographic and geospatial risks critical to global decision-making.

1. **The Need**

During a global crisis, the overabundance of unstructured data becomes a hindrance. The fundamental problem that this project solves is the inability to process and correlate complex variables in real time.
Organizations need to understand not only how many cases there are, but where they are growing the fastest, which demographics are most vulnerable, and how death rates compare between different regions. Without clear visualization, leaders operate under "data blindness," resulting in delayed responses, inefficient allocation of resources, and failures to communicate risks to stakeholders.

2. **Strategic Importance and Impact**

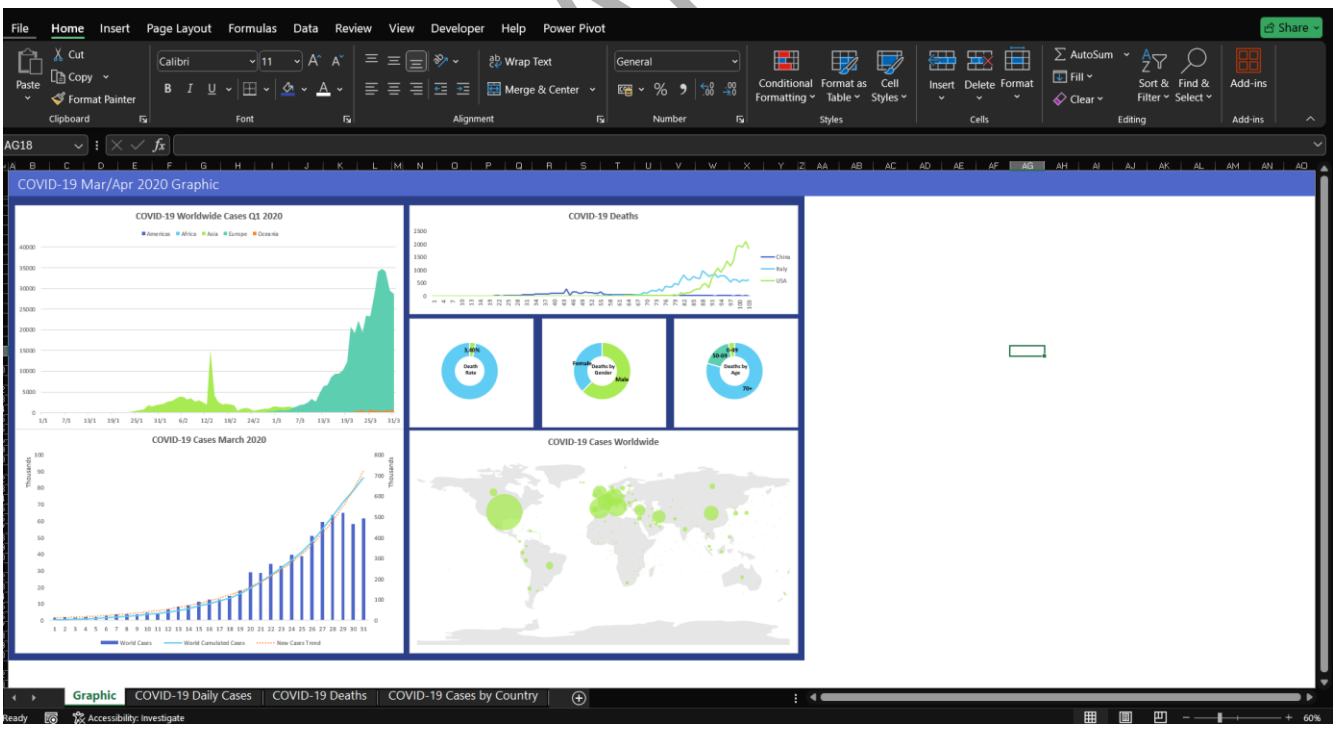
This project transforms raw data into an actionable visual narrative that impacts organizational resilience:

- **Predictive Capacity:** By integrating exponential trend lines, the system not only shows the past, but also alerts about the future trajectory of risk, allowing preventive actions before saturation peaks.
- **Multidimensional Correlation Analysis:** Through the use of secondary axes and combined graphs, the dashboard allows you to visualize the relationship between daily and cumulative growth, facilitating the identification of the "flattening curve".
- **Geospatial Intelligence:** The implementation of bubble maps based on coordinates (Latitude/Longitude) provides an immediate understanding of global infection hotspots, essential for supply chain and mobility management.
- **Vulnerability Segmentation:** Demographics (age and gender) charts allow decision-makers to design specific protection strategies for the most at-risk segments, optimizing the impact of health policies.

3. **Technology Stack and Technical Competencies**

For this project, I used advanced Business Intelligence and Data Storytelling techniques to ensure that the complexity of the data was accessible without losing technical rigor:

- **Time Series Visualization:** Use of Stacked Area Charts to visualize the regional composition of the pandemic and its chronological evolution.
- **Geospatial Modeling:** Design of custom 2D bubble charts on top of world mapping, dynamically scaling the size of data points to represent the magnitude of cases.
- **Trend Analytics:** Implementation of exponential regressions and data smoothing for the interpretation of accelerated growth.
- **Interface Design (UX for Data):** Creation of a centralized "Graphic Sheet" with standardized dimensions, optimized legends and clear visual hierarchy to facilitate the consumption of information.
- **KPI Structuring:** Breakdown of fatality metrics using Donut Charts interconnected with critical demographic variables.



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Arroyo Tanta

Project 8. Automotive Intelligence Hub: High-Impact Visualization Architecture for Automotive Retail Management

In this project I designed a Business Intelligence ecosystem that transforms fragmented financial and business metrics into an actionable visual narrative, enabling 360° monitoring of operational performance and regional financial health in real-time.

1. The Business Need

For a vehicle retailer with a national presence (such as in Australia), geographic dispersion and product hierarchy (makes vs. models) often create silos of information. The critical problem this project solves is the **lack of visibility into profitability and sales composition**.
Companies need to understand not only how much they sell, but where the growth potential is, how cash flows from opening balance sheet to closing, and whether additional services (such as maintenance contracts) are meeting market penetration targets. This project eliminates the complexity of static tables to provide immediate answers to address questions.

2. Strategic Importance and Impact

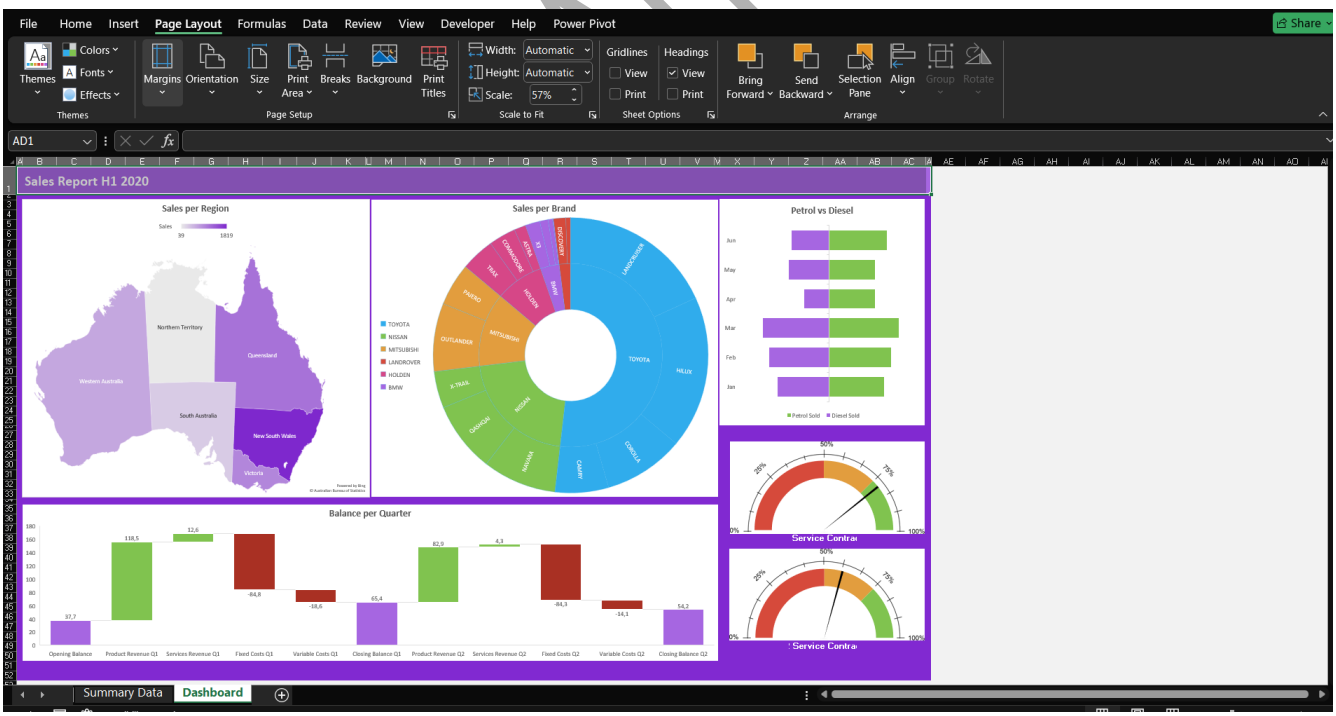
This dashboard directly impacts four pillars of decision-making:

- **Inventory and Portfolio Optimization:** By using Sunburst charts, management can instantly visualize which brand and model combinations dominate the market, enabling stock decisions based on actual demand ratios.
- **Cash Flow Transparency:** The implementation of the Waterfall chart allows you to break down the impact of operating income and expenses quarter by quarter, identifying exactly where value is gained or lost between the opening and closing balances.
- **Market Change Management (Petrol vs. Diesel):** Through a funnel/population graph, the energy transition in sales is monitored, allowing the company to adapt its commercial strategy to the change in fuel preferences.
- **Operational Efficiency Monitoring:** The design of custom Gauge charts provides an immediate visual KPI on the sale of service contracts, allowing branch managers to act on the performance of additional sales on the spot.

3. Technology Stack and Technical Competencies

To build this solution, I used advanced Data Analytics and Visual Computing techniques.

- **Geospatial Visualization Engineering:** Implementation of regional heat maps with location-specific data filtering logic.
- **Hierarchical Data Modeling:** Structuring data for multi-level visualizations (Sunburst), ensuring the integrity of the proportions.
- **Financial Analytics:** Design of waterfall charts to track capital variations and Q1/Q2 results.
- **Layered Charts Architecture:** Creation of needle gauges by overlaying and rotating series of donut and pie charts, demonstrating advanced technical customization capabilities.
- **Data Normalization and Transformation:** Processing of negative values and manipulation of scales for direct comparison visualizations (Funnel bars).



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Arroyo Tanta

Project 9. Real Estate Market Intelligence: Data Architecture for Real Estate Investment Optimization

In this project I developed a market intelligence engine that automates the analysis of auction settlement rates and trends, allowing you to identify high-yield investment opportunities in Melbourne real estate with surgical precision.

1. The Business Need

Real estate is one of the most volatile and information-saturated markets. The critical problem this project solves is "market myopia": the inability of investors and agencies to distinguish between temporary fluctuations and actual price trends at the local (suburban) level.
Without a tool that correlates property type (House vs. Unit) with location and seasonality, companies risk overpaying for assets or ignoring emerging markets. This project centralizes the fragmentation of auction data to mitigate financial risks and maximize the possible Return on Investment (ROI).

2. Strategic Importance and Impact

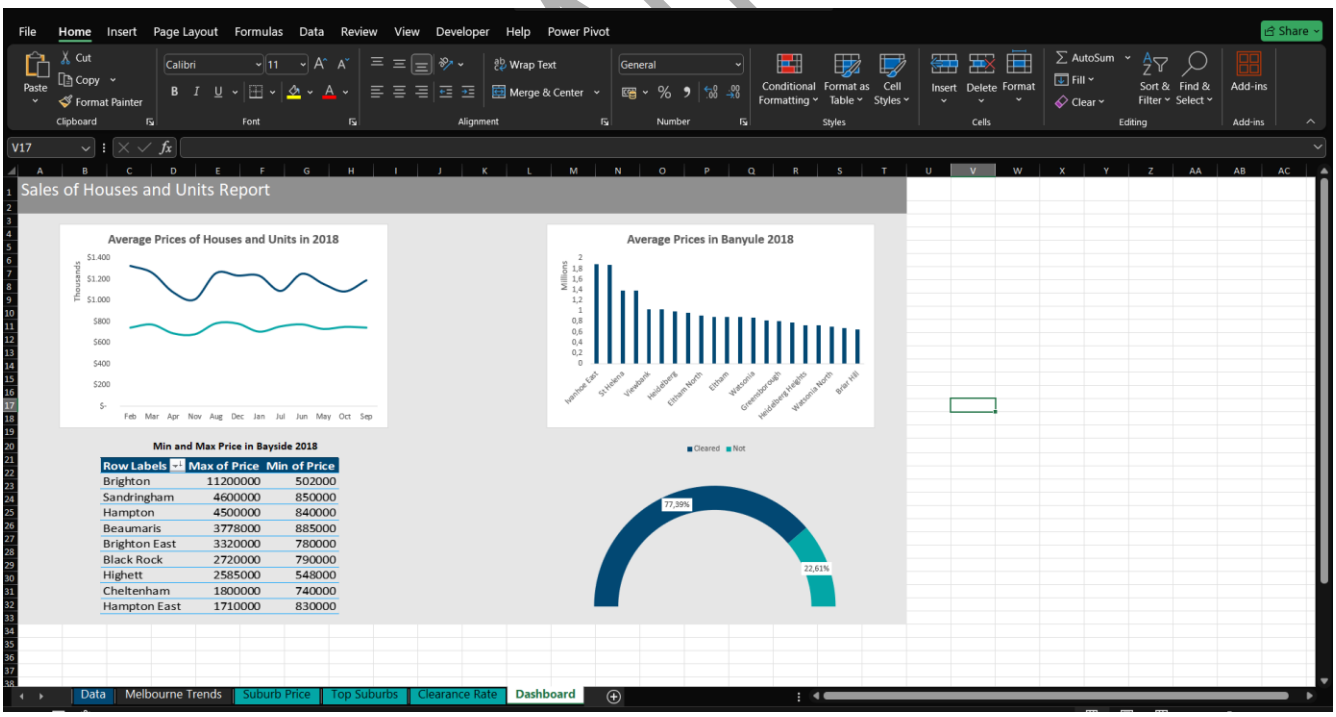
This analytics system transforms real estate management through three pillars of impact:

- **Hyper-Local Price Segmentation:** Through dynamic filtering by Council Area and month, the project allows management to compare the performance of specific suburbs (such as Banyule or Bayside), facilitating territorial expansion strategies based on real sales data.
- **Market Health Analysis (Clearance Rates):** The implementation of Gauge Charts provides a real-time "market temperature" metric. This allows you to decide if it is the optimal time to sell or if the market is favoring buyers.
- **Identifying Outliers and Market Tops:** By calculating all-time highs and lows per suburb, the system helps set auction bid limits and long-term equity projections.

3. Technology Stack and Technical Competencies

For this project, an advanced Business Intelligence (BI) stack was used within the Microsoft Excel ecosystem.

- **Data Modeling (Power Pivot):** Conversion of massive data structures into smart tables (Sales Table) to ensure the integrity and scalability of the analysis.
- **Pivot Table and Pivot Caches Engineering:** Advanced data caching management to enable independent complex calculations, such as Calculated Items, critical for custom metrics.
- **Dynamic Data Visualization:** Creation of high-fidelity PivotCharts (Line, Column and Donut), including line smoothing techniques for seasonal trend analysis.
- **Advanced Analytics:** Use of calculated field logic to create a Gauge Chart from scratch, allowing intuitive visualization of KPIs (Settlement Rates) for users.
- **UX/UI Design for Dashboards:** Application of professional visual styles, elimination of cognitive noise (gridlines and field buttons) and optimization of the information architecture in the central panel.



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Arroyo Tanta

Project 10. Market Intelligence Hub: Dynamic Dashboard Architecture for Real Estate Decision Making

In this project I designed an automated analytical command center with VBA macro integration that centralizes critical market metrics, allowing stakeholders to explore price trends and settlement rates with a single click, eliminating latency in data reporting.

1. The Business Need

In highly competitive real estate (such as Melbourne), information arrives in a fragmented and massive way. The business problem that this project solves is operational inefficiency in market analysis.

Typically, analysts waste hours cross-referencing data from different suburbs, property types, and time periods. Businesses need a "single source of truth" that is not only accurate, but allows non-technical managers to explore data without relying on an IT team, facilitating rapid responses to changes in supply and demand.

2. Strategic Importance and Impact

This dashboard isn't just a collection of charts; It is a competitive strategy tool that impacts:

- **Agility in the Investment Cycle:** Thanks to the implementation of interconnected Slicers, the user can filter by advisory area and month simultaneously, reducing the time of prospecting profitable areas.
- **Market Health Diagnosis:** By monitoring Clearance Rates, the company can predict whether the market is entering a cooling or overheating phase, optimizing sales timing.
- **Visualization of Comparative KPIs**: The use of visual indicators (such as dynamic trend arrows for unit prices) allows you to instantly identify whether an asset is appreciating above or below the regional average.
- **Data Integrity and Security:** The implementation of sheet protection and refresh automation protocols ensures that the tool is robust against human error, guaranteeing always reliable reports.

3. Technology Stack and Technical Competencies

For the construction of this solution, advanced Business Intelligence (BI) and automation techniques were used, demonstrating a deep mastery of the analytics suite:

- **Excel Business Intelligence:** Using complex PivotTables and PivotCharts as a calculation engine for multidimensional data segmentation.
- **Automation with VBA (Visual Basic for Applications):** Development of custom macros for workflow automation (Refresh All) and book lifecycle management (CloseWB), improving operational efficiency.
- **Interface Engineering (UX/UI):** Application of Linked Images for dynamic visualizations, design of tiled layouts for executive readability and customization of CSS-like styles in data segmenters.
- **Advanced Functional Logic:** Implementation of statistical formulas and custom number formats with conditional logic for trend iconography.

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Arroyo Tanta

Advanced Data Analytics & Scalable Modeling (Projects 11-14)

Project 11. Automated Retail Intelligence Hub: Scalability and Performance Analysis in a Business Chain.

In this project I designed an automated data ecosystem using Power Query that integrates multiple transactional sources, eliminating information silos and enabling real-time performance analysis for strategic decision-making.

1. Business Need

In the cosmetics distribution industry, agility is key. The main problem lay in the data disconnect: sales information arrives weekly in flat text files, while product, agent, and branch lists lived in separate, static, and cluttered spreadsheets.
This fragmentation generates an error-prone manual process, where the management team wastes days consolidating data before being able to analyze the performance of brands or the productivity of agents, points analyzed in this project. For this reason, the company needs a "Single Source of Truth" that must be able to absorb new data weekly with a single click.

2. Strategic Importance and Impact on Decision-Making

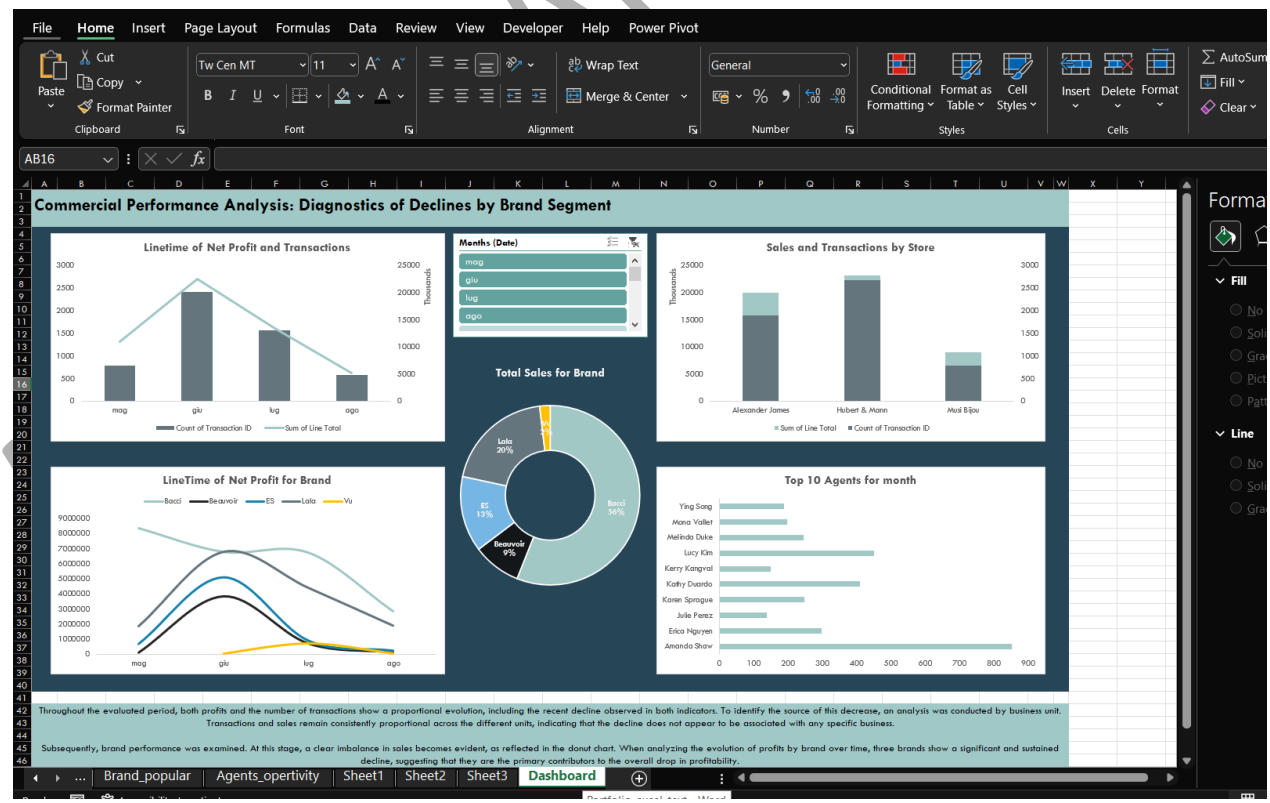
This project transforms raw data into actionable intelligence with three direct impacts:

- **Portfolio Optimization:** Allows you to instantly identify which brands (like Bacci or Lala in this case) are leading revenues, facilitating inventory and marketing decisions based on facts, not intuitions.
- **Talent Management (Performance Tracking):** By accurately quantifying the volume of items processed by each agent (identifying, for example, the 411 items managed by profiles such as Kathy Duardo), the company can implement fair incentive schemes and detect operational bottlenecks.
- **Operational Scalability:** The system is designed for the future. Thanks to the folder architecture, the system is automatically updated when new files are added, reducing report preparation time by 90%.

3. Technology Stack: Data Engineering with Power Query

To ensure the robustness and cleanliness of the process, I used a stack focused on ETL (Extract, Transform, Load) Automation:

- **Power Query (M-Language):** The main engine. I used it to normalize agent names, perform Fill Down brand hierarchies, and clean up inconsistent barcodes.
- **Multi-Source Transformation:** Implemented complex Joins (combinations) between TXT files and Excel tables to enrich transactions with store and product metadata.
- **Dynamic Data** Modeling: Creation of custom revenue calculation columns and logical filters to remove duplicate or corrupt records (Data Cleansing).
- **Business Intelligence (PivotTables):** Design of dynamic reports that allow management to segment the reality of the business by brand, agent and week in an interactive way.



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Arroyo Tanta

Project 12. Sales Performance Optimizer: Data Architecture for Goal Control and Consumption Trend Analysis

In this project, I developed a data transformation engine (ETL) that automates the normalization of complex sales records, allowing to identify performance gaps against targets and product seasonality with full operational accuracy.

1. The Business Need

The retail and distribution industry is constantly faced with the challenge of "trapped data": financial reports designed to be read, but not analyzed. The critical problem that this project solves is the **inflexibility of the source data**, where structures with empty rows, irregular headers, and horizontally arranged data (months as columns) prevent any dynamic analysis.

Without an architecture that normalizes these records, companies suffer from "analytics paralysis," wasting hours on manual cleaning and being unable to calculate vital metrics such as **quota compliance percentage** or historical trends. This project centralizes and structures information to mitigate reporting errors and maximize revenue stream visibility.

2. Strategic Importance and Impact

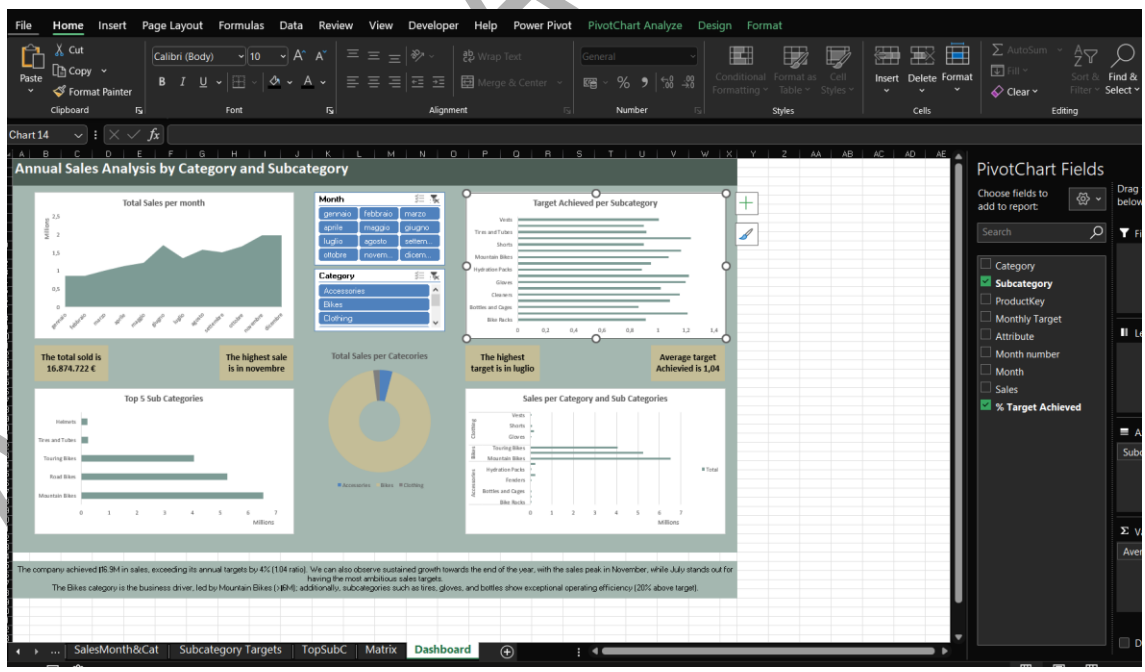
This analytics system transforms commercial management through three pillars of impact:

- **Business Effectiveness Monitoring (% Target Achieved):** By creating compliance metrics, the project allows management to instantly identify which subcategories are exceeding their targets and which require immediate intervention, optimizing resource allocation.
- **Seasonality and Time Flow Analysis:** By transforming month names into numerical and chronological dimensions, the system makes it easy to read trends through **Stacked Area Charts**, allowing you to predict peaks in demand in key categories such as *Accessories* and *Clothes*.
- **Data Scalability and Automation:** Implementing a cleansing pipeline ensures that any new data is automatically integrated into the model. This eliminates the risk of human error in spreadsheet manipulation and ensures the integrity of monthly reports.

3. Technology Stack and Technical Competencies

For this project, a Business Intelligence (BI) stack focused on data engineering was used within the Microsoft Excel and Power Query ecosystem.

- **Data Engineering with M Language (Power Query):** Implementation of transformation scripts for deep data cleansing, including **Fill Down** logic for categories and the removal of structural noise (redundant rows and columns).
- **Normalization and Unpivoting:** Conversion of horizontal matrices into a tabular model of **Fact Table**, fundamental to enable the use of PivotTable calculation engines.
- **Transformation of Temporal Dimensions:** Use of logic in M Language for the conversion of text labels to numerical values of the month, allowing a precise chronological ordering in time series visualizations.
- **Calculation of Business KPIs:** Creation of custom columns for the analysis of goal fulfillment (Sales / Monthly Target), formatted for direct executive interpretation.
- **Dynamic Trend Visualization:** Design of **Pivot Charts** of stacked areas with advanced segmentation by categories, optimizing the information architecture to highlight the performance of February and other critical periods.



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Arroyo Tanta

Project 13. Enterprise Sales Intelligence: Multi-Dimensional Data Architecture and Predictive Analytics with DAX

In this project I designed a BI ecosystem based on relational models and DAX functions, which allowed the transition from flat pivot tables to high-impact OLAP visualizations for year-over-year growth analysis, after optimizing the efficient loading of data using Power Query.

1. The Business Need

In corporate environments with large volumes of data, traditional analytics tools fail to correlate multiple sources (Sales, Goals, Products, Staff). The critical issue is the fragmentation of the metric: the difficulty of comparing current performance against previous periods in a dynamic and hierarchical way. Without a robust data model, companies cannot perform drill-down analysis without losing the integrity of the information. This project solves the need to centralize complex business rules in a single truth model, eliminating information silos and enabling a 360° view of business performance.

2. Strategic Importance and Impact

This analytics system elevates decision-making through three advanced capabilities:

- **Time Intelligence:** Using DAX features such as SAMEPERIODLASTYEAR, management can evaluate organic growth year-over-year (YoY), identifying whether the increase in sales is a real trend or pure seasonality.
- **Hierarchical Information Navigation:** The creation of Hierarchies allows users to navigate from the macro category to the specific product in a single click, optimizing response time to deviations in sales.
- **Advanced Visualization and OLAP Flexibility:** By converting pivot tables into OLAP formulas (CUBE functions), the rigidity of standard layouts is broken, allowing you to create freely designed dashboards with Sunburst charts linked to Slicers, improving the user experience (UX) for distribution analysis.

3. Technology Stack and Technical Competencies

For this project, a high-level Business Intelligence workflow was implemented:

- **Relational Data Modeling (Power Pivot):** Establishing relationships (1:N) between fact tables and dimensions, creating a star schema that guarantees the referential integrity and performance of the calculation engine.
- **Metrics Engineering with DAX:** Development of complex measures using CALCULATE to modify filter contexts and temporal intelligence functions for advanced comparative analysis.
- **OLAP Architecture and Cube Formulas:** Conversion of the dynamic model into member and cube value functions, allowing you to design unconventional visualizations (such as the Sunburst chart) that maintain full interactivity with Slicers.
- **Metadata Enrichment (Hierarchies):** Technical categorization of data to facilitate the discovery of insights through hierarchical structures integrated into the data model.
- **Query Optimization (Power Query):** Efficient file loading and data pre-processing to ensure that the Power Pivot model operates with clean, correctly typed data.

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Project 14. Dynamic Fiscal Analytics: End-to-End Architecture for Revenue Control and Operational Optimization with Power BI

With this project I designed an advanced BI infrastructure that automates data ingestion and the calculation of critical KPIs, ensuring accurate financial reporting aligned with the corporate fiscal calendar and the geographical reality of the business.

1. The Business Need

Most companies encounter many difficulties in their data analysis because of "Fiscal Myopia" – the inability to align transactional data with actual financial cycles (which rarely coincide with the calendar year).
This project solves the chaos of manually cleaning data scattered across customers, orders, and staff. Without a structure that normalizes these dimensions and resolves geographical ambiguities, management makes decisions based on fragmented data, which leads to errors in the allocation of budgets and sales goals.

2. Strategic Importance and Impact

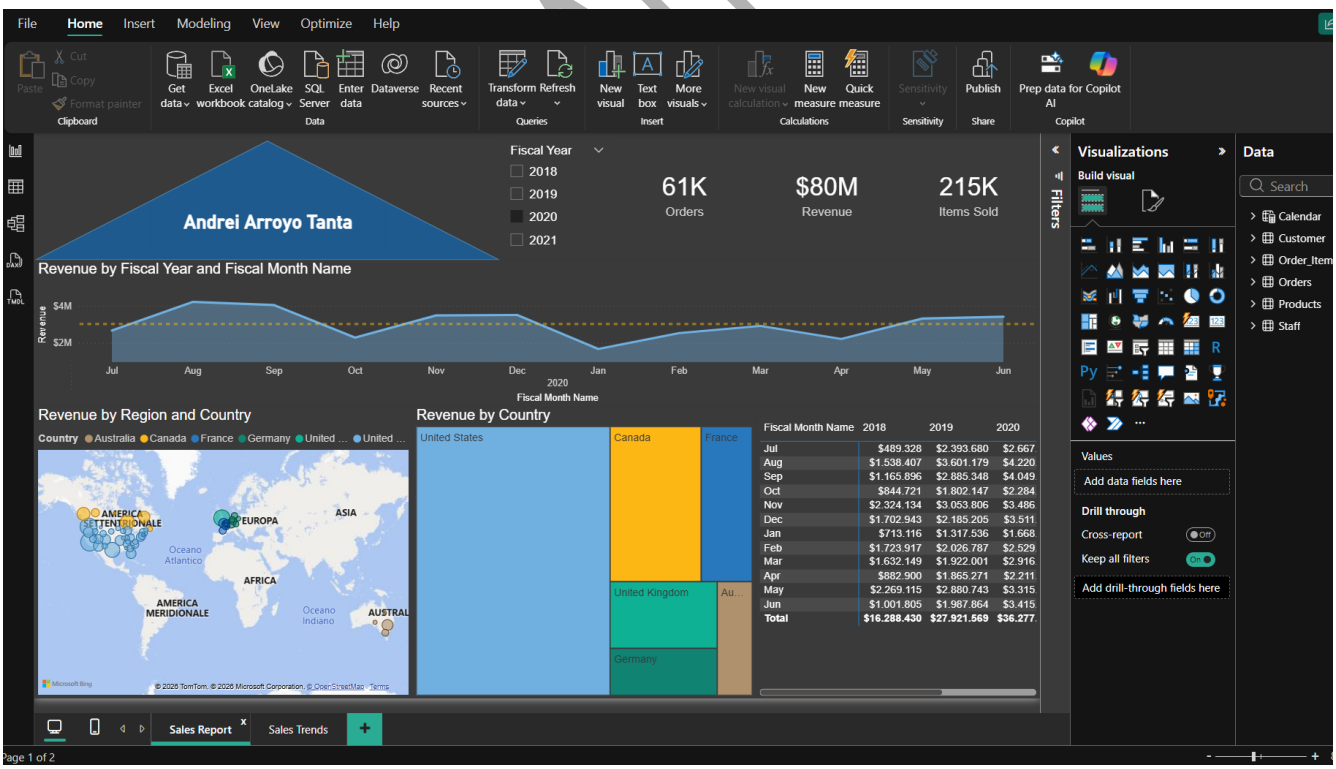
This analytics ecosystem transforms the operation through three pillars of impact:

- **Comparative Temporal Intelligence:** Thanks to the implementation of Time Intelligence functions, the company can visualize Month-to-Month (MoM) growth and percentage variations with absolute accuracy. Using Waterfall Charts allows you to identify exactly how much revenue was boosted or slowed down.
- **Hierarchical and Drill-Down Navigation:** Creating fiscal and geographic hierarchies allows leaders to go from a global revenue view to a specific SKU or county analysis in seconds, reducing the time to diagnose problems instantly.
- **Data-Driven Culture with Professional UX**: By integrating automatic conditional formats, cognitive fatigue is eliminated, allowing even non-technical users to identify risk trends (red alerts) and opportunities (green arrows) instantly.

3. Technology Stack and Technical Competencies

For this enterprise-grade deployment, I used a robust stack focused on integrity and scalability:

- Advanced ETL (Power Query): Raw data cleansing and transformation, geo-disambiguation using logical concatenation, and dynamic filtering of personnel dimensions.
- Multidimensional Modeling (Star Schema, Power Pivot and DAX): Design of a relational model (1:N) with a Dynamic Calendar Table created in DAX, configured to respond to specific fiscal cycles and dynamic dates (Max Date + 60 days).
- DAX Engineering (Data Analysis Expressions): Used for complex temporal intelligence calculations (PREVIOUSMONTH, SAMEPERIODLASTYEAR).
    - Use of DIVIDE with error handling and CALCULATE for revenue segmentation.
- Visualization and UI/UX: * Implementation of synchronized Slicers and corporate hex color themes.
    - Use of third-party visuals for advanced distribution analytics.



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